

BEYOND DIRECT IMPORT SOURCES

A True-Source Dependency Assessment of U.S. Gallium Supply Before and After China's Export Controls

Blessing Ameh

Independent Researcher

Working Paper - Version 1.1 - October 2026

Not peer reviewed

Abstract

Supplier diversification is often used as an indicator of supply-chain resilience, yet the country recorded as a direct importer's supplier may not be the country that produced or processed the material. This working paper develops and applies the True-Source Dependency Assessment (TSDA), a quantitative framework for testing whether changes in direct U.S. suppliers correspond to deeper reductions in upstream foreign dependence. The first application examines U.S. gallium-metal imports (HTS 8112.92.1000) using a pooled pre-shock baseline of 2019-2022 and a post-shock period of 2023-2025. Direct-supplier concentration increased materially: the HHI rose from 1,557.5 to 2,416.9, while the dominant supplier share rose from 25.65% to 40.01%. China's recorded direct share fell from 18.04% to 8.67%, but its annualized direct shipment quantity changed little. At the same time, primary-production concentration increased from an HHI of 9,498.0 to 9,751.3 under the principal USGS-based series, with China's share of named primary production rising from 97.44% to 98.74%. Intermediary tracing shows that major alternative routes through Canada, Germany, Japan, Taiwan, and Slovakia cannot all be treated as independent true sources because refining, recycling, transformation, and mixed-code limitations leave part of their U.S.-bound upstream provenance unresolved. The gallium case therefore does not establish a deeper structural reduction in U.S. foreign dependence during 2023-2025. The paper also demonstrates why transparent unresolved classifications, sensitivity analysis, and shock validation are necessary when public trade data cannot identify shipment-level provenance.

Keywords: critical minerals; gallium; supply-chain resilience; supplier diversification; import dependence; HHI; trade networks; China export controls; true-source dependency

1. Introduction

Critical-mineral supply chains create a measurement problem for resilience policy. A country can appear to have diversified its suppliers because it buys from a larger or different set of trading partners, while still relying on the same upstream country for production, refining, or another essential processing stage. In that setting, direct-import diversification and true-source diversification are not the same thing.

Gallium provides a useful test case. The United States is fully net-import reliant for gallium, while global primary low-purity production is overwhelmingly concentrated in China. China introduced export controls on gallium- and germanium-related items in July 2023, effective August 1, 2023, and tightened controls on relevant dual-use exports to the United States in December 2024. These events create an observable supply shock against which changes in U.S. sourcing can be examined.

Existing research already shows why a single-stage view can be incomplete. Cheng et al. (2024) demonstrate that combined vulnerability across multiple supply-chain stages can be substantially larger than vulnerability measured at individual stages. Brainard, Gulley, and Nassar (2026) show that direct trade partners can function as intermediaries, so direct trade statistics may misstate the producing-country provenance behind imports. Heydari et al. (2025), in a systematic review of critical-mineral supply-chain resilience research, likewise identify persistent gaps in midstream and downstream analysis.

This paper develops and applies a complementary question: when U.S. sourcing changes after a disruption, did the country's underlying foreign dependence actually decline, or did the visible supplier relationship change while the upstream structure remained concentrated? The True-Source Dependency Assessment (TSDA) addresses this question by comparing the direct U.S. supplier layer with production, processing, and intermediary evidence, and by requiring uncertainty to remain unresolved where public data cannot support a defensible attribution.

The contribution is therefore methodological as much as descriptive. TSDA does not create new customs or production statistics. It converts fragmented trade, production, processing, and supplier information into a structured test of whether apparent diversification represents a genuine reduction in underlying foreign dependence. Gallium is the initial proof of concept; germanium is reserved for a later extension using the same documented framework.

2. Literature and Measurement Gap

2.1 Diversification is not equivalent to resilience

Diversification can reduce exposure to a single immediate supplier, but the resilience effect depends on whether the added routes are genuinely independent. If several direct suppliers obtain feedstock from the same upstream producer or processor, the number of suppliers can increase without eliminating the common bottleneck. The relevant analytical object is therefore not only the count or concentration of direct suppliers, but the relationship between visible trade routes and the upstream stages that make those routes possible.

2.2 Multi-stage vulnerability

Cheng et al. (2024) quantify vulnerability across multiple stages of electric-vehicle battery supply chains and find that combined vulnerability can be materially larger than vulnerabilities measured at individual steps. Although their application concerns battery materials rather than gallium, the analytical implication generalizes: resilience can be overstated when stages are assessed separately and a common upstream dependency is not carried through the supply chain.

2.3 Intermediaries and provenance

Brainard et al. (2026) directly address provenance in mineral trade. Their approach constructs mineral trade networks and traces direct and indirect pathways back to original producers, showing that the country recorded as an import partner can be an intermediary rather than the producing source. TSDA builds on this provenance principle but asks a different temporal question: whether the underlying structure of dependence changes when the United States appears to diversify after a shock.

2.4 Economic relevance of disruption

Nassar et al. (2024) estimate that a complete restriction of China's net gallium exports could reduce U.S. GDP by approximately \$3.1 billion, and a simultaneous complete restriction of Chinese gallium and germanium net exports could reduce U.S. GDP by approximately \$3.4 billion under their modeled assumptions. These estimates do not validate TSDA, but they establish why correctly measuring unresolved gallium dependence matters economically: the consequences of mineral disruption can propagate into semiconductor manufacturing and downstream industries whose output value greatly exceeds the value of the raw mineral itself.

3. True-Source Dependency Assessment (TSDA)

TSDA evaluates four connected components: (1) direct U.S. suppliers, (2) production sources, (3) processing or refining sources, and (4) intermediary relationships between the direct supplier and upstream sources. The central research question for the gallium application is whether changes in direct U.S. suppliers from 2019-2022 to 2023-2025 were accompanied by a robust decline in the upstream foreign concentration that could still affect U.S. access.

3.1 Study period and direct-supplier measures

The principal baseline is pooled 2019-2022, immediately preceding the 2023 export controls. The post-shock period is pooled 2023-2025. Physical quantity is the principal trade-flow measure where units are comparable; customs value is retained for unit-value analysis.

For supplier *i*, pooled supplier share is the supplier's summed import quantity over the period divided by total U.S. import quantity over the same period. Concentration is measured using the Herfindahl-Hirschman Index (HHI), calculated as the sum of squared country shares expressed in percentage points. The dominant-country share is reported alongside HHI. Annual HHIs are retained as a sensitivity series so the pooled-period result does not conceal year-to-year volatility.

3.2 Upstream tracing

Intermediary tracing is conducted for each supplier accounting for at least 10% of U.S. imports in either comparison period, plus additional suppliers needed to cover at least 90% of pooled U.S. import quantity. A 5% threshold is used as a sensitivity screen. National trade statistics, USGS production information, Eurostat Comext, Statistics Canada, Japan Customs/JOGMEC evidence, Taiwan customs data, and other authoritative sources are used where available. A direct supplier is not automatically treated as the true source when evidence indicates refining, recycling, transformation, or imported feedstock.

A key restriction is that a refiner's aggregate import shares are not mechanically converted into the upstream composition of its U.S.-bound exports. Transformation can break the one-to-one relationship between imports and onward exports, and public trade data generally do not identify which incoming material fed a particular outgoing shipment. Mixed customs codes are treated as contextual evidence rather than mineral-specific weights when the target mineral cannot be isolated.

3.3 Measuring robust change

TSDA treats a decline in concentration as robust only when the principal specification shows a decline and the direction remains negative under the specified sensitivity analyses. The initial gallium application exposed a wording ambiguity in the original 'No Diversification' classification because the pooled direct-supplier HHI increased while the simple average of annual HHIs declined slightly. A dated Version 1.1 amendment therefore clarified the rule: 'No Diversification' applies where direct U.S. supplier concentration does not show a robust decline under the methodology's robustness requirement. The underlying Section 5 robustness rule was not changed.

3.4 Empirical validation

Pre-shock route classifications are recorded before the complete post-shock outcome is evaluated. A route classified as materially dependent on the restricted upstream source is expected to show at least one of declining quantity, rising unit value, or exit after the restriction. A route classified as independent is expected to remain comparatively stable or expand as sourcing adjusts. Routes with unresolved pre-shock provenance are not forced into a binary validation test; their post-shock behavior is reported as exploratory. Contradictions are retained, not relabeled as confirmation.

4. Data

The direct U.S. gallium series uses USITC DataWeb for HTS 8112.92.1000 (gallium metal). Primary-production estimates are drawn from successive U.S. Geological Survey Mineral Commodity Summaries. For the principal production series, each historical year uses the latest revised USGS country value available as of the research cutoff; a contemporaneous or first-published series is retained as sensitivity. Intermediary branches use the most detailed available national or regional statistics, with code limitations documented separately.

4.1 External consistency check against USGS

As an independent consistency check, the USITC series closely reproduces the U.S. Geological Survey's published gallium statistics. The USITC data yield 11,438 kg of gallium-metal imports in 2023, compared with 11,400 kg reported by USGS after rounding, and the calculated 2023 unit value rounds to \$365 per kilogram, exactly matching the USGS value. The independently pooled 2021-2024 USITC supplier shares also round to Canada 28%, Japan 22%, China 18%, and Germany 16%, the same source shares reported in the 2026 Mineral Commodity Summaries (U.S. Geological Survey, 2026). This agreement provides an external check that the direct-import extraction and aggregation used in the TSDA analysis are consistent with the federal summary series.

Table 1. Pooled U.S. direct gallium supplier shares

Supplier	2019-2022 share	2023-2025 share	Change
Canada	8.53%	40.01%	+31.48 pp
Germany	18.83%	21.19%	+2.36 pp
Japan	25.65%	15.02%	-10.63 pp
China	18.04%	8.67%	-9.37 pp
Taiwan	8.88%	6.10%	-2.78 pp
Other countries	20.07%	9.02%	-11.05 pp

Source: author calculations from USITC DataWeb, HTS 8112.92.1000. Shares use pooled physical quantities.

Total U.S. gallium-metal imports increased from 30,412 kg in 2019-2022 to 46,503 kg in 2023-2025. The decline in China's percentage share must therefore be interpreted with the expanding denominator in mind. China's annualized direct shipment quantity changed only from approximately 1,371 kg per year in the baseline to 1,344 kg per year post-shock, a decline of about 2.0%, even as its share of total U.S. imports fell by more than nine percentage points.

5. Results

5.1 Direct-supplier concentration increased

The principal pooled result does not show diversification at the direct-supplier stage. Direct-supplier HHI rose from 1,557.5 in 2019-2022 to 2,416.9 in 2023-2025, an increase of 859.4 points. The dominant supplier also changed from Japan, with 25.65% of pooled baseline quantity, to Canada, with 40.01% of pooled post-shock quantity. Thus, the identity of the leading supplier changed while overall direct-supplier concentration increased.

Annual HHIs were volatile: 1,794 (2019), 4,184 (2020), 3,002 (2021), 2,250 (2022), 2,162 (2023), 3,421 (2024), and 2,340 (2025). The simple average of annual HHIs fell from 2,807.5 to 2,641.0, but this does not overturn the pooled result because the sign differs across reasonable specifications. Under the Version 1.1 robustness rule, the evidence does not establish a robust decline in direct-supplier concentration.

Table 2. Core TSDA gallium concentration measures

Measure	2019-2022	2023-2025	Change / interpretation
U.S. import quantity	30,412 kg	46,503 kg	+52.91%
Direct-supplier HHI	1,557.5	2,416.9	+859.4
Dominant direct supplier	Japan 25.65%	Canada 40.01%	Dominant share +14.36 pp
Direct China share	18.04%	8.67%	-9.37 pp
Production HHI - principal	9,498.0	9,751.3	+253.3
China share of named primary production	97.44%	98.74%	+1.30 pp
Production HHI - sensitivity	9,449.4	9,730.8	+281.4
U.S. net import reliance	100%	100%	Unchanged

Production HHIs are calculated from named-country USGS production quantities. 2025 HHI is normalized to the named-country total because the USGS world total is rounded.

5.2 Upstream production became more concentrated

Primary low-purity gallium production was already highly concentrated before the shock and became more concentrated afterward. Under the principal revised series, pooled production HHI increased from 9,498.0 to 9,751.3, while China's share of named production increased from 97.44% to 98.74%. The first-published sensitivity series points in the same direction, with HHI increasing from 9,449.4 to 9,730.8. This result is important because a decline in China's direct U.S. import share did not coincide with a decline in production-stage concentration.

5.3 Processing-stage concentration remains unresolved

Country-level refining and processing shares were not reported consistently enough across the relevant countries and years to support a defensible processing-stage HHI. TSDA therefore reports the processing stage as unresolved rather than converting qualitative evidence on refining locations into unsupported quantitative concentration shares.

5.4 Intermediary tracing does not support treating major alternative routes as automatically independent

Table 3. Intermediary-tracing summary

Route	Upstream evidence	TSDA treatment
Canada	Secondary/refining source; mixed Canadian customs code; scrap/feedstock provenance unresolved.	Do not proportionally attribute Canadian U.S.-bound gallium to China or another source without transaction-level support.
Germany	Gallium-specific Eurostat data show major China exposure and a large Slovakia branch; Germany also refines/transforms material.	U.S.-bound upstream provenance unresolved; aggregate import shares are not used as export-feedstock weights.
Japan	Domestic primary production, high-purity refining, recycling, and material China-linked imports coexist.	Mixed origin; exact China content of Japan-to-U.S. gallium unresolved.
Taiwan	Most detailed national line is a mixed residual code rather than gallium-specific.	Partner shares are contextual only; gallium-specific provenance unresolved.
Slovakia	High-purity refining and secondary recovery are identified; low-purity/recycled feedstock origin remains incomplete.	Included in 5% sensitivity route review; not treated as a clean independent source.
Russia	USGS identifies domestic primary production materially larger than the U.S.-bound route.	Preserved as Russia in conservative attribution scenarios, without claiming shipment-level proof.

The principal 90% tracing target is met by Canada, Germany, Japan, China, and Taiwan, which together account for 90.98% of pooled post-shock U.S. import quantity. Slovakia is additionally traced for the 5% sensitivity threshold.

Because several large routes remain unresolved, an exact true-source HHI is not forced. Instead, two conservative attribution scenarios are reported. Scenario A leaves unresolved branches separate at the direct-supplier level. Scenario B attributes all unresolved non-Russia branches to China while retaining Russia as Russia. The baseline-to-post HHI rises from 1,557.5 to 2,416.9 under Scenario A and from 8,886.3 to 9,819.3 under Scenario B. These are scenario endpoints, not formal identification bounds. Their value is directional: both increase despite sharply different assumptions about unresolved provenance.

6. Validation Against the Post-Shock Response

The validation stage is intentionally conservative. Only routes that were sufficiently classified using pre-shock evidence receive a formal prediction. Germany, Canada, Japan, Taiwan, and Slovakia remain exploratory because their U.S.-bound pre-shock upstream provenance could not be resolved well enough for a clean binary test. China is the restricted upstream source itself and is used as a reference route rather than as an intermediary prediction.

Table 4. Post-shock route behavior and validation treatment

Route	Annualized quantity change	Unit-value change	Relative to market	Validation status
China	-2.0%	+14.6%	-33%	Reference route; no intermediary prediction
Germany	+129.4%	+149.9%	+45%	Exploratory; pre-shock provenance unresolved

Route	Annualized quantity change	Unit-value change	Relative to market	Validation status
Canada	+856.3%	+81.3%	+5%	Exploratory; scrap/feedstock provenance unresolved
Japan	+19.4%	+73.0%	+1%	Exploratory; mixed upstream sources
Taiwan	+40.0%	+254.8%	+106%	Exploratory; mixed customs code
Russia	-68.6%	+30.7%	-24%	Formal independent-route prediction contradicted, but materially confounded
Slovakia	+218.5%	+15.6%	-33%	5% sensitivity route; exploratory
All U.S. imports	+103.9%	+72.0%	0%	Market benchmark

Unit value equals customs value divided by quantity. "Relative to market" compares each route's unit-value change with the approximately 72% market-wide increase: $(1 + \text{route change}) / (1 + \text{market change}) - 1$. Changes are descriptive and do not establish causation.

Relative to the market-wide increase, direct Chinese gallium became about 33% cheaper on a relative basis, while Germany (+45%) and Taiwan (+106%) became relatively more expensive; Canada (+5%) and Japan (+1%) moved broadly with the market, and Russia (-24%) and Slovakia (-33%) became relatively cheaper. The pattern is descriptive, not causal.

Russia is the only route that met the pre-shock condition for a formal independent-source test. Its annualized U.S. quantity contracted by approximately 68.6%, which contradicts the prediction that an independent route should remain comparatively stable or expand. However, this is not a clean falsification. Public Law 117-110 suspended normal trade relations with Russia and Belarus effective April 9, 2022, making HTS column 2 rates applicable to Russian goods. For gallium under HTS 8112.92.1000, the relevant rate changed from 3% under column 1-general treatment to 25% under column 2. The 22-percentage-point tariff increase is a material route-specific confounder for the observed post-2022 contraction.

A second validation limitation concerns prices. Market-wide U.S. gallium unit value increased by about 72% between the pooled periods. Under the original validation rule, a rising route unit value counts as one form of post-shock pressure, but when the whole market becomes more expensive this condition loses discriminatory power. Market-relative unit-value changes are therefore retained as exploratory evidence only; the original pre-specified rule is not rewritten after observing the gallium result.

7. Discussion

7.1 A lower direct China share is not the same as lower China dependence

The most important empirical lesson is the distinction between share and structure. China's direct share of U.S. gallium-metal imports fell from 18.04% to 8.67%, yet annualized direct quantity from China was nearly flat and the total U.S. market expanded. At the same time, the U.S. became more concentrated among direct suppliers and global primary production became even more concentrated in China. A direct-share decline can therefore coexist with persistent or increasing upstream dependence.

7.2 The direct supplier can change while the bottleneck remains upstream

Canada emerged as the dominant direct U.S. supplier after the shock, while Germany also increased its share. Those changes are economically meaningful, but they cannot automatically be interpreted as independent source creation. Canadian supply includes secondary and refining activity with unresolved scrap/feedstock provenance. Germany is a refining and trading node with substantial China-linked import exposure and a major Slovakia relationship, but transformation prevents a defensible mechanical allocation of its aggregate import mix to its U.S.-bound exports. The correct conclusion from public data is therefore narrower: visible sourcing changed, but the evidence does not establish that the underlying upstream bottleneck changed to the same extent.

7.3 Why unresolved is a substantive result

In mineral supply-chain analysis, unresolved provenance is not simply missing paperwork. It identifies a measurement boundary that can affect resilience conclusions. If public data cannot establish whether a new direct supplier represents domestic production, recycling, imported feedstock, or transformed material from the original concentrated source, then treating that route as independent would create false precision. TSDA therefore treats 'unresolved' as an explicit analytical outcome and uses scenarios or qualitative evidence where precise attribution would exceed the data.

7.4 Economic and policy implications

The economic value of a true-source measure is decision quality. If diversification is judged only by the countries appearing in customs records, firms and public agencies can mistake a new route for a new source and direct resources toward arrangements that preserve the same upstream bottleneck. A stage-specific dependency assessment can instead help distinguish where additional domestic production, refining, recycling, allied sourcing, stockpiling, or supplier development may still be needed. It can also provide evidence when diversification has genuinely reduced concentration, avoiding unnecessary intervention where the underlying structure has already improved.

For gallium specifically, the current evidence supports caution rather than a claim that U.S. sourcing became structurally independent after the 2023 controls. The direct supplier mix changed, but direct concentration did not robustly decline, primary production became more concentrated in China, U.S. net import reliance remained 100%, and the provenance of several major alternative routes remains partly unresolved.

The policy relevance is reinforced by Section 844 of the National Defense Authorization Act for Fiscal Year 2026, which adds gallium and germanium to the covered materials under the Department of Defense sourcing restrictions in 10 U.S.C. § 4872, effective two years after enactment, and extends the recovered-material exception to those minerals. The provision makes source and processing provenance directly relevant to future defense procurement eligibility (Pub. L. 119-60, § 844).

8. Limitations

First, TSDA relies primarily on public trade and government production data. These sources are not shipment-level supply-chain records and generally cannot reveal which incoming batch of feedstock was used in a particular U.S.-bound export after refining or transformation.

Second, some national tariff classifications combine gallium with other metals. Mixed codes can indicate trading relationships but cannot be treated as precise gallium quantities without a defensible separation method.

Third, primary low-purity production concentration is not identical to the structure of high-purity refined gallium or recycled supply. The production-stage HHI therefore measures an important upstream layer, not the entire gallium value chain.

Fourth, HHI captures geographic concentration but not all dimensions of resilience. It does not directly measure inventory, product quality, contract terms, capacity utilization, substitution, logistics constraints, or the speed with which alternative capacity can scale.

Fifth, the 2023-2025 period contains other policy and commercial shocks. The Russia tariff episode demonstrates why a route's post-shock contraction cannot automatically be attributed to China's gallium restrictions. In addition, USGS reports that China lifted its U.S.-specific gallium export ban for one year in November 2025, after the August 2023 licensing controls and December 2024 ban. Because this easing occurred within the final year of the post-shock window, 2025 import quantities and prices reflect both adjustment under restriction and a late-year policy relaxation, which complicates causal interpretation of the pooled 2023-2025 outcome.

Finally, this is an initial single-mineral application. Germanium and later mineral cases are needed to determine how well the TSDA classification and validation logic generalize across commodities with different data structures and supply-chain architectures.

9. Conclusion

This paper develops TSDA as a method for asking a narrower and more decision-relevant question than whether the United States buys a critical mineral from more countries: whether the underlying foreign dependence behind those visible suppliers actually changed.

The initial gallium application does not show a robust reduction in direct-supplier concentration from 2019-2022 to 2023-2025. Direct-supplier HHI increased, the dominant supplier share increased, primary-production concentration increased under both principal and sensitivity specifications, and U.S. net import reliance remained at 100%. Although China's recorded direct share fell substantially, its annualized direct shipment quantity was nearly unchanged, and major alternative supply routes cannot all be treated as independent true sources from the public evidence available.

These findings do not establish that every post-shock U.S. gallium shipment ultimately depended on China. They establish something more defensible: the available evidence does not support a conclusion that the United States achieved a deeper structural reduction in gallium dependence during the study period. That distinction is the purpose of TSDA. The next research stage will apply the same documented method to germanium and then compare the two cases to evaluate the framework's broader performance.

Data and Reproducibility Statement

The analysis is based on public-source trade and government mineral data. The versioned TSDA methodology, methodological amendment, gallium analysis workbook, source inventory, transformation notes, and validation record are being assembled for public release with the working paper. A permanent repository identifier should be added to this section when the public release is created. No proprietary company data are used in the analysis.

Declaration

The author reports no financial conflict of interest related to this working paper. This version is a preprint/working paper and has not yet undergone journal peer review.

References

- Brainard, J. L., Gulley, A. L., & Nassar, N. T. (2026). Indirect mineral import reliance and provenance. *Mineral Economics*. <https://doi.org/10.1007/s13563-026-00629-8>
- Cheng, A. L., Fuchs, E. R. H., Karplus, V. J., et al. (2024). Electric vehicle battery chemistry affects supply chain disruption vulnerabilities. *Nature Communications*, 15, 2143. <https://doi.org/10.1038/s41467-024-46418-1>
- Heydari, M., Mitchell, P., Cullen, L., Andrieu, B., Cabrera Serrenho, A., & Cullen, J. (2025). A systematic review of resilience in the critical minerals supply chains, needed for the low-carbon energy transition. *Renewable and Sustainable Energy Transition*, 100127. <https://doi.org/10.1016/j.rset.2025.100127>
- Ministry of Commerce of the People's Republic of China & General Administration of Customs. (2023, July 3). Announcement No. 23 of 2023 on export controls for gallium- and germanium-related items. <https://policy.mofcom.gov.cn/claw/clawContent.shtml?id=97101>
- Ministry of Commerce of the People's Republic of China. (2024, December 3). Notice 2024 No. 46: Strengthening controls on exports of relevant dual-use items to the United States. <https://cset.georgetown.edu/publication/china-rare-earth-export-ban/>
- Nassar, N. T., Shojaeddini, E., Alonso, E., Jaskula, B., & Tolcin, A. (2024). Quantifying potential effects of China's gallium and germanium export restrictions on the U.S. economy (Open-File Report 2024-1057). U.S. Geological Survey. <https://doi.org/10.3133/ofr20241057>
- U.S. Congress. (2022). Suspending Normal Trade Relations with Russia and Belarus Act, Pub. L. No. 117-110. <https://www.govinfo.gov/app/details/PLAW-117publ110>
- U.S. Congress. (2025). National Defense Authorization Act for Fiscal Year 2026, Pub. L. No. 119-60, § 844, 139 Stat. 976. <https://www.govinfo.gov/app/details/PLAW-119publ60>
- U.S. Geological Survey. (2020-2026). Mineral Commodity Summaries: Gallium, annual editions used for historical production series and revisions. <https://www.usgs.gov/centers/national-minerals-information-center/gallium-statistics-and-information>
- U.S. Geological Survey. (2026). Mineral Commodity Summaries 2026: Gallium. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-gallium.pdf>
- U.S. International Trade Commission. DataWeb: U.S. trade and tariff data, HTS 8112.92.1000 (Gallium metal). <https://dataweb.usitc.gov/>
- Eurostat. International trade in goods - Comext database. <https://ec.europa.eu/eurostat/web/international-trade-in-goods/database>
- Statistics Canada. Canadian International Merchandise Trade Web Application. <https://www150.statcan.gc.ca/n1/pub/71-607-x/71-607-x2021004-eng.htm>
- Japan Customs, Ministry of Finance. Trade Statistics of Japan - Code Lists and Data Downloads. https://www.customs.go.jp/toukei/sankou/code/code_e.htm